

Introduction

- ❖ A water cooler is a convenient appliance that provides quick access to clean drinking water.
- ❖ It serves as a central hub in homes and offices to keep people healthy and hydrated.
- ❖ Modern units often purify tap water to remove contaminants and improve taste.
- ❖ Use large 5-gallon plastic jugs placed on top or inside the bottom.

Purposes of Water Cooler Dispenser

Aim

- ❖ Design a compressor-based water cooler dispenser that efficiently delivers clean, safe, and continuously cooled drinking water.

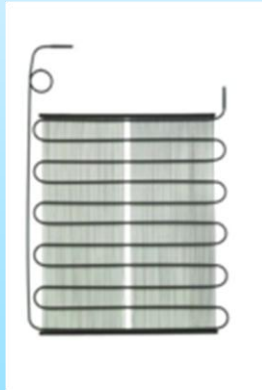
Objectives

- ❖ Apply vapour compression refrigeration for water cooling.
- ❖ Build a working prototype with suitable materials.
- ❖ Evaluate cooling efficiency, energy use and temperature stability.

Conclusion

- ❖ Successfully built a compressor-based water cooler dispenser.
- ❖ Delivers clean, cool water using vapor compression refrigeration.
- ❖ Safe, energy-efficient, and easy to operate.
- ❖ Future improvements: solar power, auto temperature control, enhanced insulation.
- ❖ The standard temperature range for a water cooler is 5°C to 10°C.

Components of Water Cooler Dispenser



**Technological University
(Mandalay)
Department of Mechanical
Engineering**

“Water Cooler Dispenser”



Compressor

- ❖ A compressor is a device that operates mechanically to increase the pressure of a gas through volume reduction.

Condenser

- ❖ The condenser is a vital component in a compressor-based water cooler dispenser, responsible for releasing heat absorbed by the refrigerant during the cooling process.

Capillary Tube

- ❖ The capillary tube is one of the essential expansion devices that are found in small refrigeration systems, for instance, water cooler dispenser that are based on compressors.

Evaporator Coil

- ❖ The evaporator coil is one of the main components of a water cooler dispenser. Its main function is to absorb heat from the water and make the water cold.

Working Procedure of Water Cooler Dispenser

- ❖ Water enters the storage tank.
- ❖ The compressor compresses the refrigerant and sends it to the condenser.
- ❖ The condenser removes heat from the Refrigerant
- ❖ The refrigerant passes through the capillary tube, reducing its pressure and temperature
- ❖ The cold refrigerant flows through the evaporator coil and absorbs heat from the water.
- ❖ The cold water is stored in the tank and dispensed through the outlet when needed.

Components Assembly and Frame Setting

